Bernulli & Binomial (random variables X)

A Bernulli Ran. Var. X can take two values:

Range $= \{0, 1\}$

Does not happen — Does happen

$P(X=0) = 1 - p = q$

$P(X=1) = p$

If n independent Bermoulli random trails are performed $(B_1, B_2, \dots, B_n)$

Let X = total # successes = $B_1, B_2, \dots, B_n$

X = binomial$(n, p) \Rightarrow P(X=K) = \binom{n}{K} p^K q^{n-K}$

Ex $n = 2$

Failure

$P(X=0) = P(F_1 F_2) = P(F_1) P(F_2) = q^2$

Success

$P(X=1) = P(S_1 F_2) + P(F_1 S_2) =$

$= P(S_1) P(F_2) + P(F_1) P(S_2) =$

$= pq + qp = 2pq$

$P(X=2) = P(S_1, S_2) = P(S_1) P(S_2) = p^2$

$$\boxed{q^2 + 2qp + p^2 = (q+p)^2 = 1}$$

Ex: Prob. of getting 3 sixes on 12 dices rolls

$X \sim$ binomial$(12, 1/6)$ $\quad P(X=3) = \binom{12}{3} \cdot (1/6)^3 (5/6)^9$

Large $n \to \infty$

$X \sim$ Binomial$(n, p) \to X \sim$ normal$(np, \sqrt{npq})$ if npq not too small

$\to X \sim$ poisson(λ) with $\lambda = np$ if np or nq small